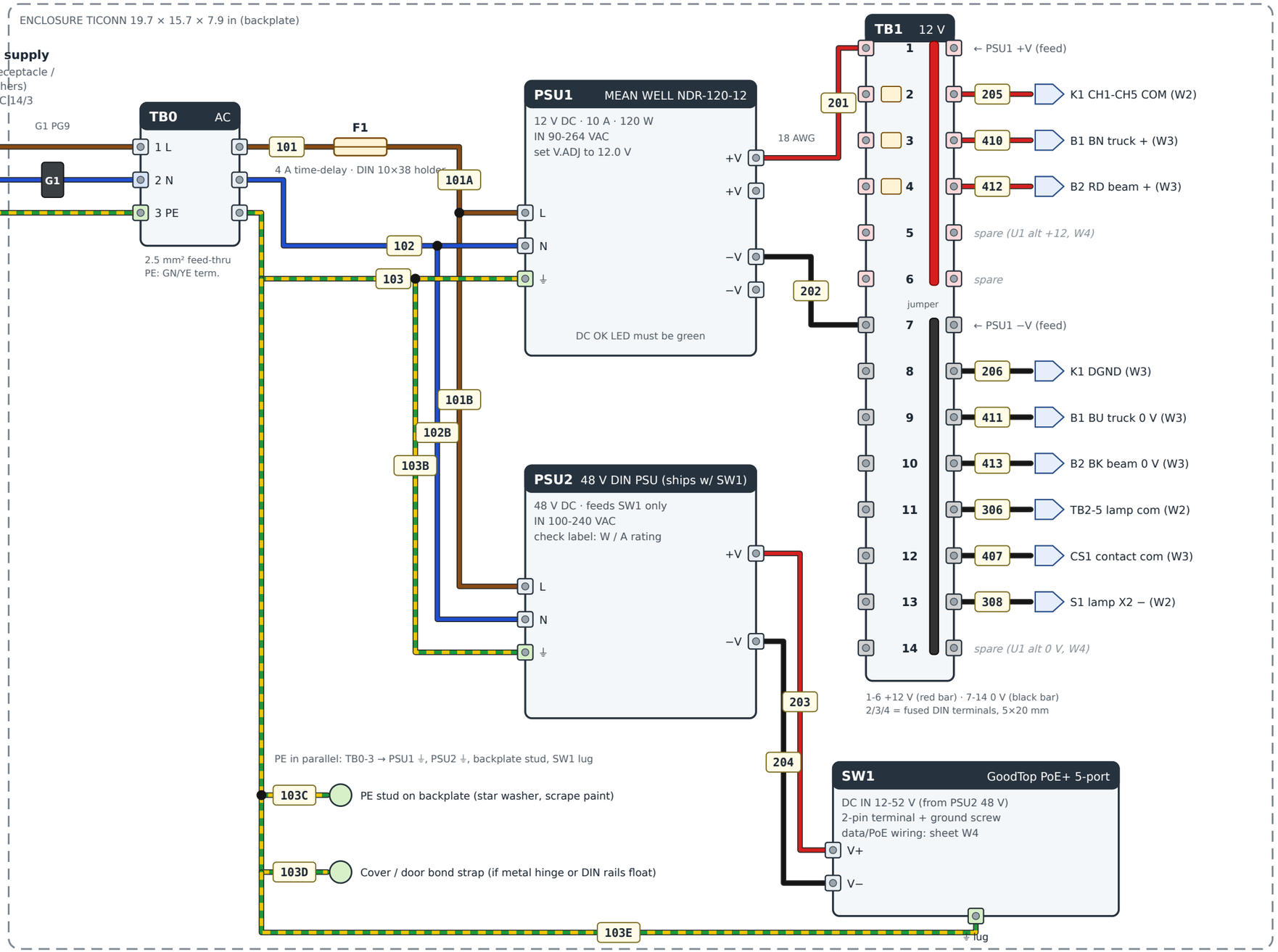
W1 · Power distribution

120 VAC in through G1, one fuse, two DIN supplies. K1 is PoE-powered: leave its 7-36 V input EMPTY.



Power budget (per box)

PSU1 12 V (120 W available)
2 stack lights, 1 lamp + buzzer each ≈ 6 W
S1 pilot lamp (12 V LED) ≈ 0.3 W
B1 laser + B2 beam ≈ 2 W
total ≈ 8-11 W (<10 %)
U1 reader if moved to PSU1 (W4 option) +36 W max
PSU2 48 V → SW1
SW1 + PoE: SPL1/U1 ≤ 25.5 W, K1 ≈ 3-5 W → check label
F1 (120 VAC)
4 A time-delay (Class CC/midget 10×38). NDR-120 inrush ≈ 30 A cold start: do NOT use fast-acting.
If PSU2 label > 1.5 A in, go 5 A.
F2-F4 (12 V branch, 5×20 mm in fused terminals)
F2 2 A K1 relay COM bus · F3/F4 0.5 A sensors
Protects 18-22 AWG field cable from the 10 A supply.

Power-up check (before K1/U1 connected)

1. Meter TB0-1 → TB0-2 ≈ 120 VAC, TB0-1 → TB0-3 ≈ 120 VAC.
2. PE continuity TB0-3 ↔ backplate stud ↔ PSU cases < 1 Ω.
3. Close F1. PSU1 DC OK lit; TB1-1 → TB1-7 = 12.0 V (trim V.ADJ).
4. TB1-2/3/4 → TB1-7 = 12 V (fuses in).
5. PSU2 +V → −V = 48 V; SW1 power LED on.
6. K1 7-36 V terminal: nothing landed. PoE only.
7. TB1 0 V is NOT bonded to PE (floating DC system).

Wire & termination spec

AC (101-103): 18 AWG MTW/UL1015 600 V, BN L · BU N · GN/YE PE
DC 12 V / 48 V: 18 AWG UL1007, RD + · BK 0 V
All stranded ends: insulated ferrules (18 AWG = 1.0 mm²)
Two wires in one terminal: twin ferrule (PSU1 L, N, ⏚)
Torque: PSU 4.4 lb·in · DIN terminals 5 lb·in (check part)
Label both ends of every wire with its number

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

RFID PANEL BOX · RPB-001 · WIRING DIAGRAM Gielow Pickles · G5 Croswell · one box per dock door	REV **9 (proposed)**	SHEET **1 of 5**
W1 AC + DC POWER DISTRIBUTION G1 → TB0 → F1 → PSU1 / PSU2 → TB1 12 V bus, SW1 48 V	DATE Oct 28, 2026	CHECKED _____

W2 · K1 relay outputs → stack lights, buzzer, S1 lamp

Relay commons are jumpered to +12 V (fused F2). Relays switch +12 V out on NO. Tower common returns to 0 V.

TB1-2 +12 V via F2 2 A (W1)

K1 Waveshare ESP32-S3-POE-ETH-8DI-8RO

relay section · 10 A 250 VAC / 30 VDC
TCA9554 @0x20, bit n = CHn+1

CH1	RED lamp	NO · C · NC
CH2	AMBER lamp	NO · C · NC
CH3	GREEN lamp	NO · C · NC
CH4	BUZZER	NO · C · NC
CH5	S1 pilot lamp	NO · C · NC
CH6	spare	NO · C · NC
CH7	spare	NO · C · NC
CH8	spare	NO · C · NC

205 jumpers
C1→C2→C3→C4→C5
18 AWG RD, ferruled

205 · 301 · 205 · 302 · 303 · 304 · 305 · 306

TB2 LAMPS

4-level / quattro
DIN terminals

1 RED · 2 AMBER · 3 GREEN · 4 BUZZ · 5 COM 0V

306

TB1-11 0 V (W1)

Panel wiring K1→TB2: 18 AWG, color as shown.

S1 LOAD COMPLETE lamp (green)

22 mm illuminated PB, 12 V LED
X1 + / X2 − contacts: W3

G7 · X1 + · X2 − · 308

TB1-13 0 V (W1) · in cable W-CS (6-cond), see W3

HL1 12 V LED tower R/A/G + buzzer

5-wire lead · verify colors on arrival

301 · G5 · 306

RED · AMBER · GREEN · BUZZER · common

Cable W-HL1: tower lead or 18/5 SJOOW to G5

HL2 12 V LED tower R/A/G + buzzer

5-wire lead · verify colors on arrival

301 · G6 · 306

RED · AMBER · GREEN · BUZZER · common

Cable W-HL2: tower lead or 18/5 SJOOW to G6
Lead too short → splice in IP67 JB at mast, keep colors

Terminal order on the real board: VERIFY (pluggable 3-pin per CH, silkscreen NO / C / NC). Relay LEDs mirror coil state.

Stack-light notes

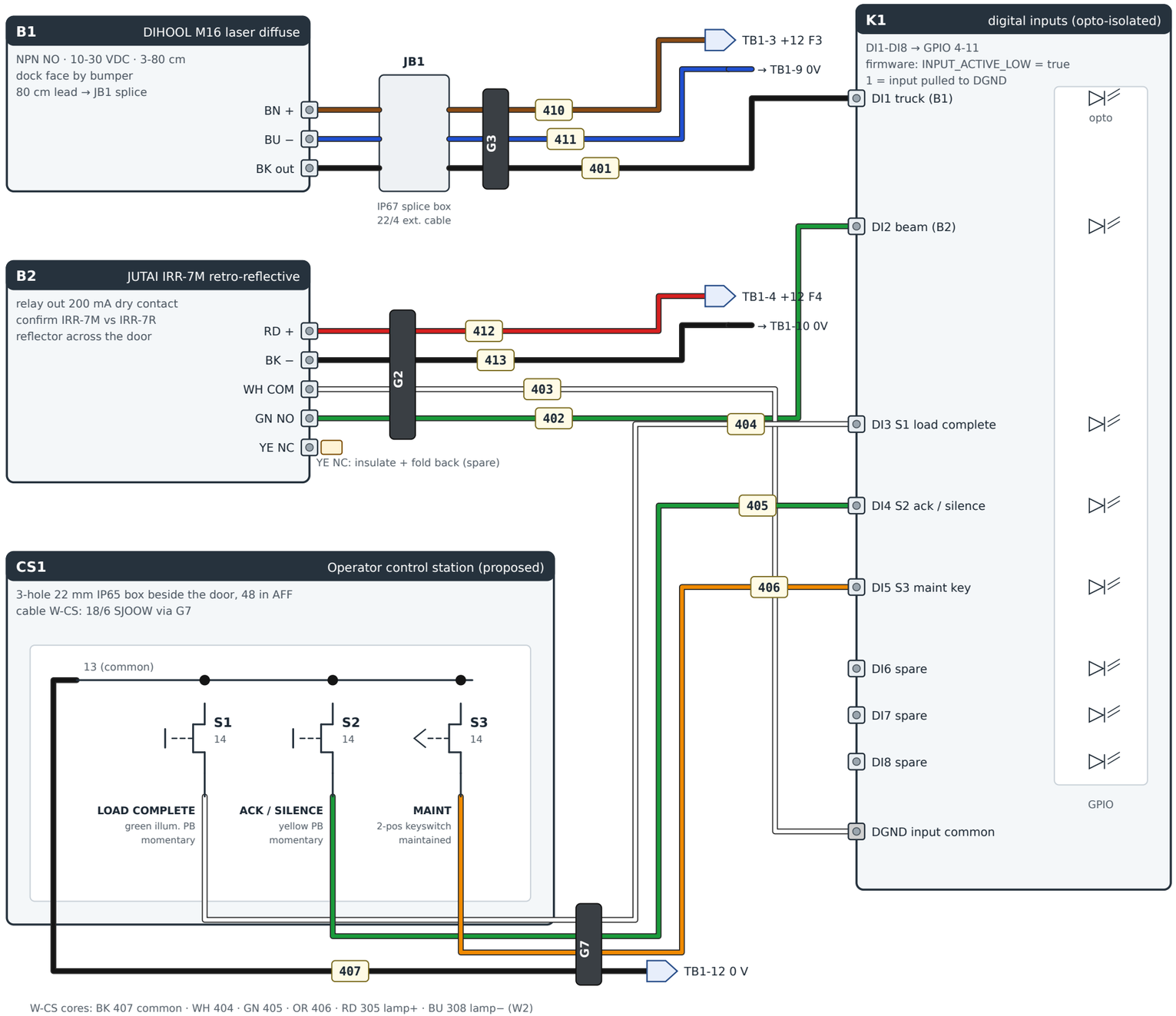
Lamp truth: firmware sets exactly ONE of CH1/CH2/CH3. CH4 = buzzer, CH5 = S1 lamp.
HL1 + HL2 in parallel at TB2 (one wire in, two out per level). 4-level terminals,
or 2 stacked 2-level terminals with a bridge. Each tower ≈ 3 W → F2 2 A is plenty.
Tower common polarity: meter it. If the common must be +12 V (common-anode tower):
move 205 to TB1 0 V and 306 to TB1 +12 V. Nothing else changes.
Typical lead colors: RD red · YE amber · GN green · WH or BN buzzer · BK common.
Relay NC contacts unused: leave empty. Don't wire anything to NC on CH1-CH3.

RFID PANEL BOX · RPB-001 · WIRING DIAGRAM Gielow Pickles · G5 Croswell · one box per dock door	REV **9 (proposed)**	SHEET **2 of 5**
W2 K1 RELAY OUTPUTS CH1-CH4 → TB2 → HL1 + HL2 stack lights · CH5 → S1 lamp · CH6-8 spare	DATE Oct 28, 2026	CHECKED _____

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

W3 · K1 digital inputs → sensors + operator controls

Opto inputs switch when DIn is pulled to DGND (NPN sink / dry contact). DGND bonded to TB1 0 V.



W-CS cores: BK 407 common · WH 404 · GN 405 · OR 406 · RD 305 lamp+ · BU 308 lamp− (W2)

Input map

Input	Device	Closed / 1 means	Wire
DI1	B1 BK (NPN NO)	truck present	401
DI2	B2 GN NO / WH COM	beam broken*	402/403
DI3	S1 13-14	load complete pressed	404
DI4	S2 13-14	ack / silence pressed	405
DI5	S3 13-14	maintenance ON	406
DI6-8	—	spare (dry contact to DGND)	—
DGND	TB1-8	bonded to 0 V	206

Why it fails safe

B1 is NPN normally-open: cut cable / dead sensor reads 'no truck' → server shows RED.
*B2 relay: pick NO or NC on the bench so 'beam broken' = 1. With NO + light-on,
a cut cable reads 'beam clear' → reader stays idle (no false green).
S3 maint is maintained; key removable only in NORMAL.
These inputs are NOT a safety circuit. Don't use them for machine guarding / E-stop.

Bench test (README) — do this before mounting

1. DGND→TB1 0 V landed (206). Hand in front of B1 → serial 'DI1 -> 1'.
2. Block B2 → 'DI2' toggles. Move GN→YE (NC) if it reads backwards.
3. Press S1, S2; turn S3 → DI3/DI4/DI5 → 1, inputs JSON release/ack/maint true.
4. If every DI reads inverted, set INPUT_ACTIVE_LOW false in config.h.
5. If a dry contact to DGND does NOT switch the opto, the board needs a sourced input: see option B below.

Option B — if K1 DI turns out to need an external voltage

Waveshare DI wiring is still UNVERIFIED (README 'known unknowns').
If DIn needs +V to switch (opto anode at DIn, cathode at DGND):
· DGND stays on 0 V (206).
· B2 WH COM and S1-S3 common (407) move from 0 V to +12 V (TB1-5/6).
· B1 NPN can't source: swap to a PNP sensor (BK = +12 when target seen),
e.g. the PNP Heschen from PO 1811, or add a 12 V interposing relay.
Mark the change on this sheet; firmware is unchanged.

Cable

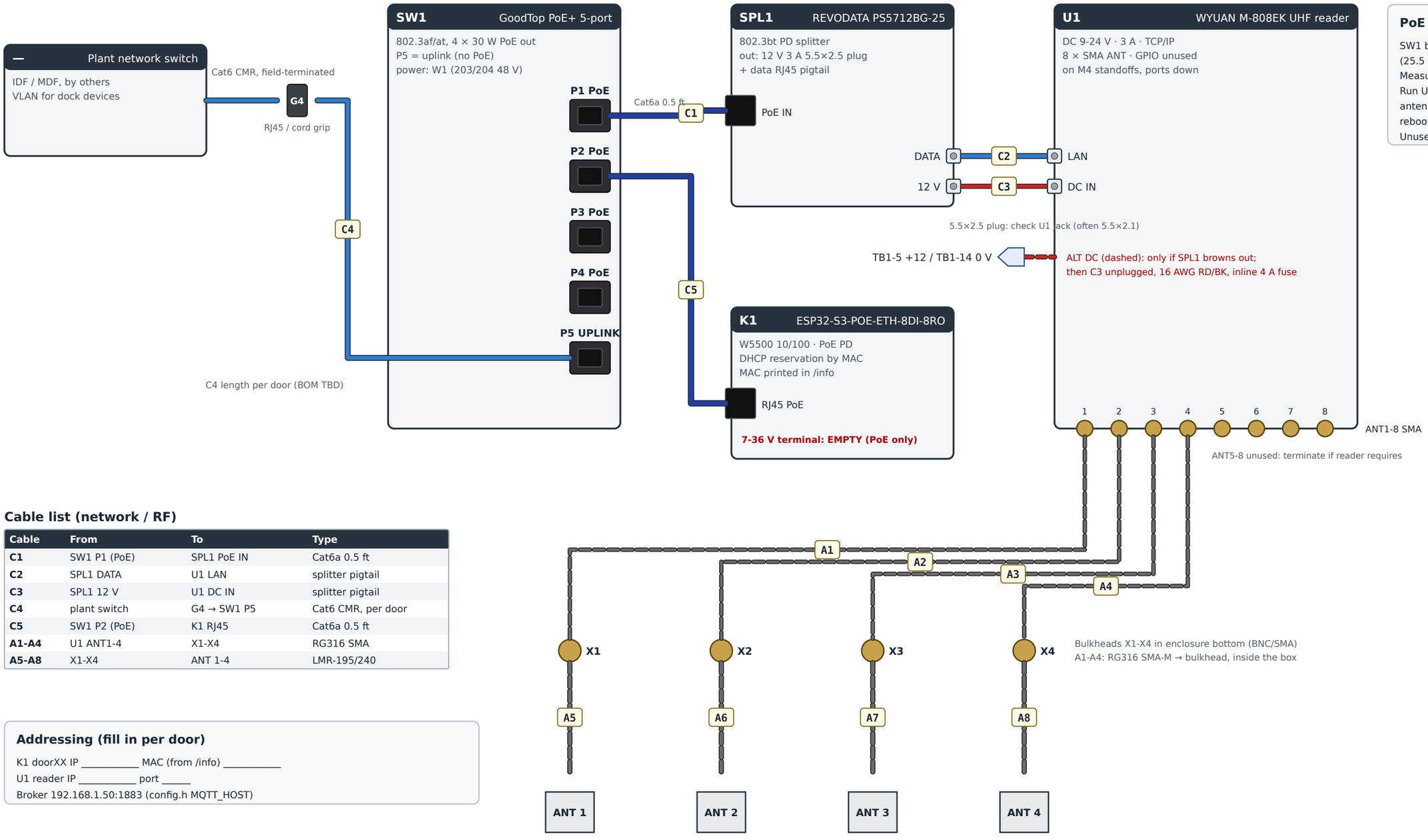
B1 ext: 22/4 PVC-jacket sensor cable, BN/BU/BK/WH(spare), ≤ 100 ft.
B2: 22/5 or the sensor's own lead. CS1: 18/6 SJOOW. Shield (if any): K1 end only.
Keep input cables away from AC (G1) and antenna coax inside the box.

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

RFID PANEL BOX · RPB-001 · WIRING DIAGRAM Gielow Pickles · G5 Croswell · one box per dock door	REV **9 (proposed)**	SHEET **3 of 5**
W3 K1 DIGITAL INPUTS B1 truck → DI1 · B2 beam → DI2 · S1/S2/S3 control station → DI3-DI5	DATE Oct 28, 2026	CHECKED _____

W4 · Network, PoE and antenna wiring

Everything is copper Ethernet + PoE inside the box. Plant network enters on C4 through G4.



PoE checks

SW1 budget ≥ SPL1 + K1 (25.5 W + ~5 W).
Measure K1 draw on bench.
Run U1 at 33 dBm, all 4 antennas, 10 min: no reboot = splitter OK.
Unused P3/P4: cap them.

Cable list (network / RF)

Cable	From	To	Type
C1	SW1 P1 (PoE)	SPL1 PoE IN	Cat6a 0.5 ft
C2	SPL1 DATA	U1 LAN	splitter pigtail
C3	SPL1 12 V	U1 DC IN	splitter pigtail
C4	plant switch	G4 → SW1 P5	Cat6 CMR, per door
C5	SW1 P2 (PoE)	K1 RJ45	Cat6a 0.5 ft
A1-A4	U1 ANT1-4	X1-X4	RG316 SMA
A5-A8	X1-X4	ANT 1-4	LMR-195/240

Addressing (fill in per door)

K1 doorXX IP _____ MAC (from /info) _____
U1 reader IP _____ port _____
Broker 192.168.1.50:1883 (config.h MQTT_HOST)

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

RFID PANEL BOX · RPB-001 · WIRING DIAGRAM Gielow Pickles · G5 Croswell · one box per dock door	REV **9 (proposed)**	SHEET **4 of 5**
W4 NETWORK · PoE · RF plant switch → G4 → SW1 → SPL1 → U1 · SW1 → K1 · U1 → 4 antennas	DATE Oct 28, 2026	CHECKED _____

W5 · Wire, terminal and gland schedules

Build from this list. Check each row off as you land it, then ring it out end to end.

Wire schedule — power + outputs

Wire	From	To	Color	AWG	Sht
101	G1 field L	TB0-1	BN	18	W1 ☐
101	TB0-1	F1 in	BN	18	W1 ☐
101A	F1 out	PSU1 L	BN	18	W1 ☐
101B	PSU1 L	PSU2 L	BN	18	W1 ☐
102	TB0-2	PSU1 N	BU	18	W1 ☐
102B	PSU1 N	PSU2 N	BU	18	W1 ☐
103	TB0-3	PSU1 ⏚	GN/YE	18	W1 ☐
103B	PSU1 ⏚	PSU2 ⏚	GN/YE	18	W1 ☐
103C	TB0-3	backplate PE stud	GN/YE	14	W1 ☐
103D	PE stud	cover bond	GN/YE	14	W1 ☐
103E	PE stud	SW1 ground lug	GN/YE	18	W1 ☐
201	PSU1 +V	TB1-1	RD	18	W1 ☐
202	PSU1 −V	TB1-7	BK	18	W1 ☐
203	PSU2 +V	SW1 V+	RD	18	W1 ☐
204	PSU2 −V	SW1 V−	BK	18	W1 ☐
205	TB1-2 (F2)	K1 CH1 C	RD	18	W2 ☐
205	K1 CH1 C	CH2 C, CH3 C, CH4 C, CH5 C	RD	18	W2 ☐
206	TB1-8	K1 DGND	BK	18	W1/W ☐
301	K1 CH1 NO	TB2-1 → HL1/HL2 red	RD	18	W2 ☐
302	K1 CH2 NO	TB2-2 → HL1/HL2 amber	OR/YE	18	W2 ☐
303	K1 CH3 NO	TB2-3 → HL1/HL2 green	GN	18	W2 ☐
304	K1 CH4 NO	TB2-4 → HL1/HL2 buzzer	WH	18	W2 ☐
305	K1 CH5 NO	G7 → S1 X1 (lamp +)	RD	18	W2 ☐
306	TB1-11	TB2-5 → HL1/HL2 common	BK	18	W2 ☐
308	TB1-13	G7 → S1 X2 (lamp −)	BU	18	W2 ☐

Wire schedule — inputs + network

Wire	From	To	Color	AWG	Sht
401	B1 BK (JB1)	G3 → K1 DI1	BK	22	W3 ☐
402	B2 GN (NO)	G2 → K1 DI2	GN	22	W3 ☐
403	B2 WH (COM)	G2 → K1 DGND	WH	22	W3 ☐
404	S1 14	G7 → K1 DI3	WH	18	W3 ☐
405	S2 14	G7 → K1 DI4	GN	18	W3 ☐
406	S3 14	G7 → K1 DI5	OR	18	W3 ☐
407	TB1-12	G7 → S1/S2/S3 13 (jumpered)	BK	18	W3 ☐
410	TB1-3 (F3)	G3 → JB1 → B1 BN	BN/RD	22	W3 ☐
411	TB1-9	G3 → JB1 → B1 BU	BU/BK	22	W3 ☐
412	TB1-4 (F4)	G2 → B2 RD	RD	22	W3 ☐
413	TB1-10	G2 → B2 BK	BK	22	W3 ☐
—	B2 YE (NC)	insulated, spare	YE	22	W3 ☐
C1	SW1 P1	SPL1 PoE IN	Cat6a	—	W4 ☐
C2	SPL1 data	U1 LAN	pigtail	—	W4 ☐
C3	SPL1 12 V	U1 DC IN	pigtail	—	W4 ☐
C4	plant switch	G4 → SW1 P5	Cat6	—	W4 ☐
C5	SW1 P2	K1 RJ45	Cat6a	—	W4 ☐
A1-4	U1 ANT1-4	X1-X4	RG316	—	W4 ☐
A5-8	X1-X4	ANT 1-4	LMR	—	W4 ☐

Terminal strips

Terminal	Function	Wires
TB0-1	L in (G1)	101 → F1
TB0-2	N in (G1)	102 → PSU1
TB0-3	PE in (G1)	103 …
TB1-1	+12 feed	201 PSU1 +V
TB1-2	+12 F2 2 A	205 K1 COMs
TB1-3	+12 F3 ½ A	410 B1
TB1-4	+12 F4 ½ A	412 B2
TB1-5	+12	spare / U1 alt
TB1-6	+12	spare
TB1-7	0 V feed	202 PSU1 −V
TB1-8	0 V	206 K1 DGND
TB1-9	0 V	411 B1
TB1-10	0 V	413 B2
TB1-11	0 V	306 TB2-5
TB1-12	0 V	407 CS1 common
TB1-13	0 V	308 S1 lamp −
TB1-14	0 V	spare / U1 alt
TB2-1	red	301 in / HL1 + HL2
TB2-2	amber	302 in / HL1 + HL2
TB2-3	green	303 in / HL1 + HL2
TB2-4	buzzer	304 in / HL1 + HL2
TB2-5	lamp com	306 in / HL1 + HL2

Glands (bottom face)

Gland	For	Cable	Size
G1	AC in 120 V	18/3 or 14/3	PG9
G2	B2 beam	22/5	PG7
G3	B1 truck (JB1)	22/4	PG7
G4	C4 uplink	Cat6	PG11 split
G5	HL1 tower	18/5	PG9
G6	HL2 tower NEW	18/5	PG9
G7	CS1 station NEW	18/6	PG9
X1-X4	antenna bulkheads	coax	BNC/SMA

Rev 9 changes vs rev 8 (proposed — mark up the DXF when approved)

1. TB0 AC entry terminals (L / N / PE) and a backplate PE stud; PE to PSU1, PSU2, SW1, cover.
2. TB1 grows to 14 positions (6 × +12 V, 8 × 0 V); TB1-2/3/4 become fused terminals F2/F3/F4.
3. TB2 lamp marshalling (5 × 4-level terminals) so HL1 and HL2 each land on their own clamp.
4. HL2 gets its own gland G6 (rev 8 shared G5 between two cables = not IP67).
5. Operator control station CS1 (S1 load complete, S2 ack/silence, S3 maint key) on K1 DI3-DI5, S1 lamp on K1 CH5. New gland G7, cable W-CS 18/6. Wires 305, 308, 404-407.
6. 48 V return gets its own number 204 (rev 8 used 203 for both legs). Sensor supplies 410-413.
7. Firmware 1.1.0: buzzer/pilot topics, blink states, server-presence fail-safe, /info fw + MAC.

Added parts for rev 9 (per box, not yet priced)

TB0: 2 × feed-thru + 1 × PE ground terminal, 2.5 mm² · TB1: 8 more terminals + 3 fused terminal holders
TB2: 5 × 4-level (quattro) terminals + end plate · jumper bars · PE stud kit
CS1: 3-hole 22 mm IP65 station, green illuminated 12 V PB, yellow PB, 2-pos keyswitch, legend plates
Cable: 18/6 SJOOW for CS1 · 18/5 for HL2 · 2 × PG9 glands (G6, G7) · fuses 2 A + 2 × 0.5 A (5×20)

Units: inches, AWG. Wire numbers are per potential: the same number on both ends and every jumper of one net. Ref RPB-001 rev 8 drawing + illustrated schematic.

RFID PANEL BOX · RPB-001 · WIRING DIAGRAM Gielow Pickles · G5 Croswell · one box per dock door	REV **9 (proposed)**	SHEET **5 of 5**
W5 WIRE · TERMINAL · GLAND SCHEDULES every conductor, from → to, with color, gauge and sheet	DATE Oct 28, 2026	CHECKED _____